Executive Business Report

Sales Forecasting & Demand Intelligence System

Executive Summary

This report presents the results of a sales forecasting and demand intelligence analysis conducted using historical retail sales data. The objective of the project was to understand past sales trends, predict future demand, identify unusual sales patterns, and segment products based on demand behavior to support better inventory planning and business decisions. The analysis shows that sales follow clear seasonal trends, with certain product categories and regions consistently outperforming others. Machine learning models were used to forecast future sales, detect abnormal sales weeks, and classify products into different demand segments. The insights from this analysis can help improve inventory planning, reduce stock shortages, minimize excess inventory, and support more informed business decisions.

Key Findings

Sales Trend Analysis

Historical sales analysis revealed that sales vary throughout the year, indicating the presence of seasonal demand patterns. Monthly and yearly trend analysis showed consistent fluctuations, suggesting that customer purchasing behavior changes during different periods of the year.

Among all product categories, **Furniture** generated the strongest projected growth based on the forecasting model. Regional analysis also highlighted noticeable differences in sales performance, indicating that inventory planning should consider regional demand instead of using a single strategy for all locations.

Forecasting Results

Several forecasting techniques were evaluated during this project. After comparing the models, **XGBoost** provided the most accurate predictions and was selected as the final forecasting model.

The model predicts the following sales trend of **Furniture** for the next three months:

Forecast Period	Predicted Sales	Confidence
Month 1	7041.121582	Expected range around forecast
Month 2	5094.810547	Expected range around forecast
Month 3	16025.856445	Expected range around forecast

Overall, the forecast indicates **increasing** demand over the next three months. Based on these results, the business should prepare inventory levels accordingly to meet expected customer demand.

Anomaly Detection Findings

Two anomaly detection techniques were applied:

- Isolation Forest
- Rolling Z-Score

Isolation Forest detected several unusual sales weeks, while the Rolling Z-Score identified statistically significant deviations from recent sales trends.

The three most significant anomalies are summarized below.

Period	Observation	Likely Business Explanation
2015-03-22	High Sales Spike	Likely caused by promotional campaigns or festive season demand.
2017-10-08	High Sales Spike	Possible festive season promotion.
2018-01-21	Low Sales Drop	Lower demand after the holiday season.

These unusual events should be reviewed regularly because they may indicate important business opportunities or operational issues.

Product Demand Segmentation

K-Means clustering was used to group product sub-categories according to their sales characteristics. Four distinct demand segments were identified.

High Volume, Stable Demand

These products consistently generate strong sales throughout the year. Maintaining sufficient inventory and frequent replenishment is recommended to avoid stock shortages.

Growing Demand

These products show increasing sales trends. Inventory levels should be gradually increased while continuously monitoring future demand.

Low Volume, High Volatility

Demand for these products changes frequently. A limited inventory with safety stock is recommended to reduce unnecessary holding costs while maintaining product availability.

Declining Demand

These products show decreasing sales over time. Inventory should be reduced, and promotional offers or clearance sales should be considered to improve inventory turnover.

Business Recommendations

1. Increase Inventory for High-Growth Products

The forecasting model predicts that the **Furniture** category will experience the strongest demand growth over the next three months. Increasing procurement and inventory for this category will reduce the risk of stock shortages and improve customer satisfaction.

Supporting Evidence: XGBoost forecast identifies Furniture as the highest-growth category.

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2. Monitor Abnormal Sales Periods More Closely

Several unusual sales weeks were detected by the anomaly detection models. Investigating these events can help identify successful marketing campaigns, seasonal demand patterns, or operational issues that may require corrective action.

Supporting Evidence: Isolation Forest detected **[replace with your count]** anomalous weeks.

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3. Implement Demand-Based Inventory Planning

Instead of managing all products using the same inventory policy, different stocking strategies should be adopted for different demand segments. Stable products should receive priority inventory allocation, while declining products should be stocked conservatively.

Supporting Evidence: K-Means clustering identified four distinct product demand segments with different sales characteristics.

Risk and Limitation

Although the forecasting system provides useful business insights, it relies primarily on historical sales data. External factors such as economic conditions, competitor activities, sudden changes in customer preferences, supply chain disruptions, and special promotional campaigns are not included in the current model. As a result, actual future sales may differ from the predicted values during unexpected business events. Regular model updates and the inclusion of additional business variables can further improve forecasting accuracy.

Conclusion

The Sales Forecasting and Demand Intelligence System demonstrates how data-driven techniques can support retail decision-making. By combining exploratory data analysis, machine learning forecasting, anomaly detection, and demand segmentation, the system provides actionable insights for inventory planning and business strategy. Implementing the recommendations from this report can help reduce inventory costs, improve product availability, and support better operational planning across the retail supply chain.